

B.Tech.(Hons.) in AGRICULTURAL AND FOOD ENGINEERING and M.Tech. in FOOD PROCESS
ENGINEERING

MICRO SPL. in ARTIFICIAL INTELLIGENCE AND APPLICATIONS, ENTREPRENEURSHIP AND INNOVATION

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	7.88 / 10
2021	CBSE Class XII	Madav Rao Scindia Public School, Shahjahanpur	97.8%
2019	CBSE Class X	Takshashila Public School, Shahjahanpur	99%

PROJECTS

Real-Time Chat Application | Self Project Dec 2025 - Present

Overview: Built a messaging platform with WebSocket communication, JWT authentication, and responsive UI serving concurrent interactions

- Engineered real-time messaging with **Socket.IO** implementing rooms, typing indicators, presence tracking achieving sub-**100ms latency**
- Implemented **JWT-based authentication** with access tokens and protected route middleware, significantly reducing unauthorized access
- Architected **PostgreSQL** backend with optimized indexing strategies for message persistence, reducing the message fetch **latency**
- Integrated **OAuth 2.0** social authentication (Google/GitHub) with seamless token exchange flow, reducing user onboarding time by **60%**

Bhakti Tradition Map | Bachelor Thesis Project | Prof. Priyadarshi Patnaik, IIT Kharagpur Aug 2025 - Nov 2025

Overview: Developed a full-stack web platform to visualize and analyze 1000+ years of Bhakti movement data using highly interactive maps

- Developed a scalable platform using **Node.js**, **Express.js**, **MongoDB**, & **JavaScript** for spatial-temporal exploration of saint networks
- Designed and implemented **RESTful APIs** with indexed MongoDB query filtering, improving query response performance by **~30–35%**
- Engineered an automated **geocoding** pipeline using OpenStreetMap (**Nominatim**), achieving **>90%** successful coordinate resolution
- Built an interactive UI using **Leaflet.js** implementing marker **clustering**, ensuring smooth performance while rendering **100s+ markers**

URL Shortener | Self Project Jun 2025 - Jul 2025

Overview: Full-stack web app with custom links, advanced click tracking, QR code generation, leveraging React, Supabase and modern tools

- Developed a responsive frontend using **React**, Tailwind CSS, and **Shadcn UI** for seamless link creation, tracking, and analytics visualization
- Implemented a serverless backend with **Supabase**, implemented authentication, URL shortening, tracking & dynamic **QR** code generation
- Designed a user **dashboard** for link management and visualizing analytics, exporting click data reducing manual tracking effort by **~40%**

Kiyo.AI | Chrome Extension | AlgoZenith Hackathon Dec 2024

Overview: Developed an AI-powered Chrome extension that enhances coding productivity by providing intelligent, context-aware assistance

- Developed and deployed a **context-aware** Chrome extension leveraging **Gemini API**, delivering AI-driven and **real-time** coding assistant
- Implemented real-time messaging updates, and **local storage** with dynamic and persistent **chat history** linked to every coding problem
- Engineered **dynamic prompt** generation and local storage, enabling **contextual**, adaptive **AI** responses and seamless user experiences

AWARDS AND ACHIEVEMENTS

- Secured '**EX**' (Excellent) grade in total of **13** courses across five semesters, consistently demonstrating academic **excellence** and sincerity
- Attained an All India Rank of **8454** in the highly competitive JEE Advanced 2022, ranking among the top **5.4%** out of 1,55,538 candidates

SKILLS AND EXPERTISE

Languages and Technologies: C, C++, Python, HTML, CSS, JavaScript, SQL, Bash, REST APIs, WebSockets, PostgreSQL, MongoDB, Supabase
Framework and Libraries, Tools, APIs: STL, React, Leaflet.js, Tailwind CSS, Node.js, Express.js, Git, Github, VS Code, Vercel, Gemini API

CERTIFICATIONS

Applied Algorithms and Data Structures | Algozenith Dec 2024 - Apr 2025

- Completed an intensive **16-week** live training program in the DSA, gaining expertise in topics such as **Linked Lists**, Binary Trees & **Graphs**
- Learnt to apply various **problem-solving** paradigms & techniques like Greedy, Divide and Conquer, **Dynamic Programming**, **STL**, Bitmask

POSITIONS OF RESPONSIBILITY

General Secretary Social and Cultural | Patel Hall of Residence | IIT Kharagpur Aug 2024 - Jun 2025

- Elected by **440+** students, led a community of **200+** including **two** secretaries for functioning of all Social and Cultural Activities of the Hall
- Negotiated a budget of **INR 2.75 Lakhs** to facilitate participation in **18** GC events conducted by **Technology Students' Gymkhana (TSG)**

Core Organising Team Member | Spring Fest | IIT Kharagpur Aug 2023 - Jun 2024

- Co-headed a team of **49** associates and **250+** campus ambassadors, expanding reach to **1200+** colleges and publishing **350+** media articles
- Successfully executed the strategic alumni contribution drives in Delhi and Uttar Pradesh raising **INR 7.52 Lakhs** through alumni network

COURSEWORK INFORMATION

MOOCs: Operating Systems | Computer Networks | Computer Organisation and Architecture | Database Management Systems (DBMS)
Computer Science: Programming and Data Structures (Theory and Laboratory) | Artificial Intelligence : Foundations and Applications
Mathematics: Linear Algebra, Numerical & Complex Analysis | Advanced Calculus | Probability and Statistics | Partial Differential Equation

EXTRA CURRICULAR ACTIVITIES

- Mentored **four** first-year students academically under the **Student Mentorship Programme**, conducted by Students' Welfare Group, TSG
- Co-directed, co-scripted and performed in the **silver-winning** Nukkad Natak '**Plastic Paradox**' for General Championship **Street Play 2025**